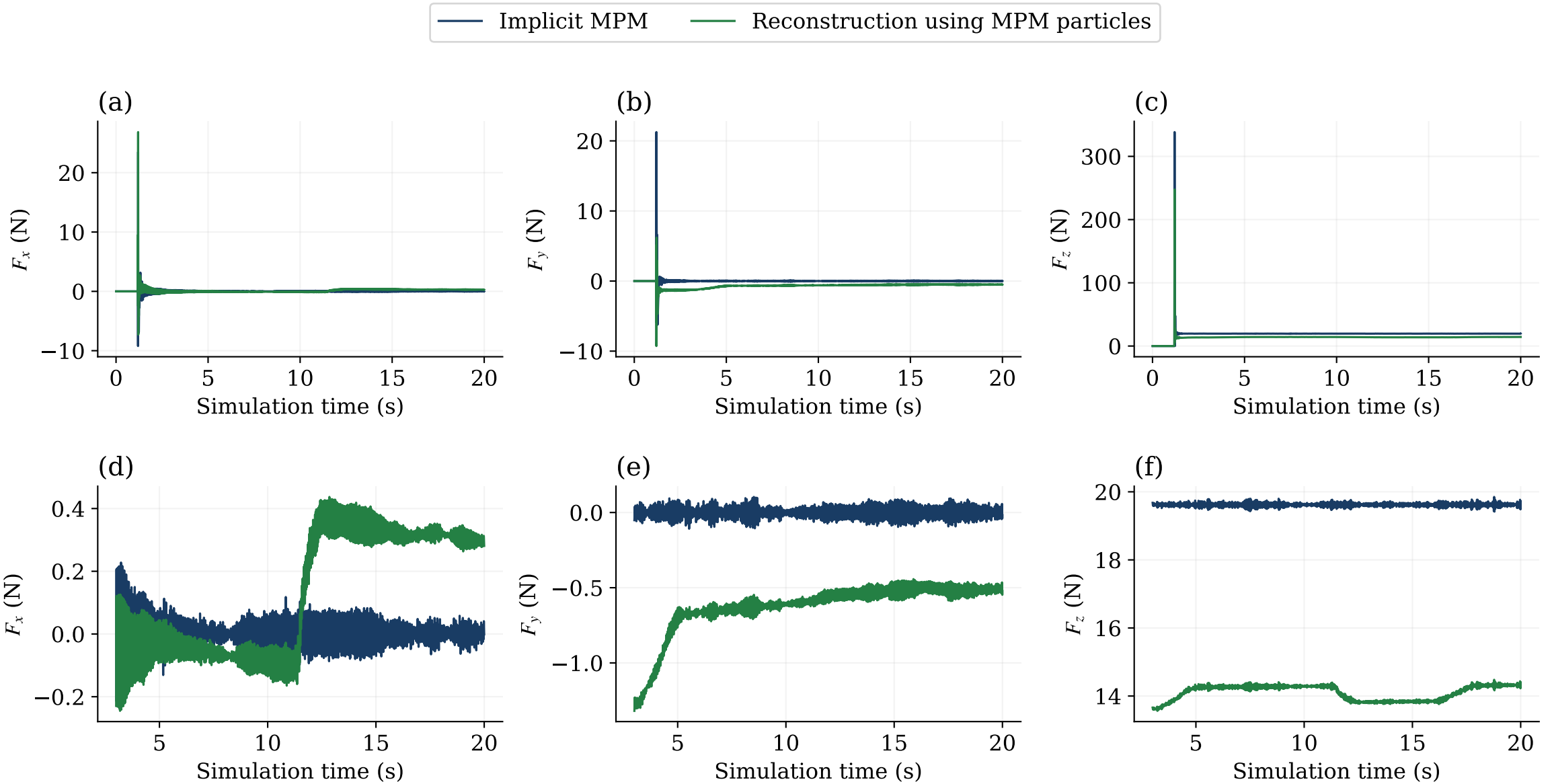



Figure 3: One paired rollout; raw world-frame material-on-object forces at 200 Hz, +z upward.
(a) Fx; (b) Fy; (c) Fz over the full recording.
(d) Fx; (e) Fy; (f) Fz from 3 s onward, with expanded force scales to show sustained support.
Recorded MPM body motion and free velocities are imposed; estimates do not feed back to the body.
The same estimator uses MPM particles and position-difference velocities; no second neural model.
Direct rigid-tray forces are excluded.